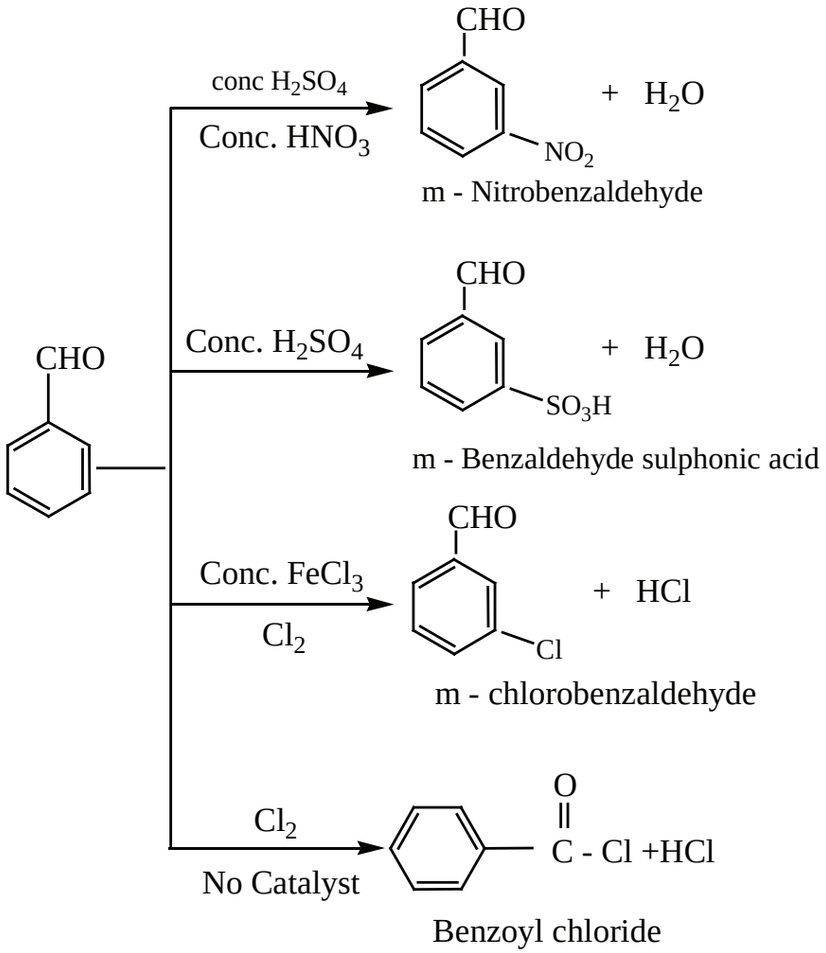
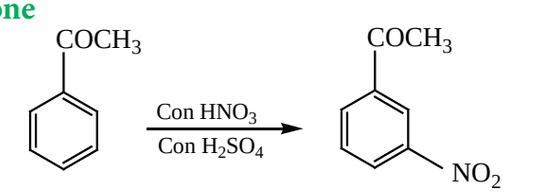
8) Electrophilic substitution reactions of benzaldehyde



Electrophilic substitution reaction of acetophenone

Acetophenone reacts with Nitrating mixture to form m – nitroacetophenone.



12.6 Test for Aldehydes

i) Tollens Reagent Test

Tollens reagent is an ammonical silver nitrate solution. When an aldehyde is warmed with Tollens reagent a bright silver mirror is produced due to the formation of silver metal. This reaction is also called **silver mirror test** for aldehydes.

$$CH_3CHO + 2\,[Ag(NH_3)_2]^+ + 3OH^- \longrightarrow CH_3COO^- + 4NH_3 + 2Ag + 2H_2O$$

Silver

ii) Fehling's solution Test

Fehling's solution is prepared by mixing equal volumes of Fehling's solution 'A' containing aqueous copper sulphate and Fehling's solution 'B' containing alkaline solution of sodium potassium tartarate (Rochelle salt)

When aldehyde is warmed with Fehling's solution deep blue colour solution is changed to red precipitate of cuprous oxide.

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